

Product Description

US Army Sustainment Command (ASC)
Packaging, Storage, and Containerization Center (PSCC)

REQUIRED CAPABILITIES

The base requirement of the contemplated contract action is for quantity one (1) environmental test chamber capable of exposing products up to cycles of temperature and humidity conditions. The chamber should meet or exceed all salient characteristics, and it should include required supporting documentation and a warranty. The Contractor would be responsible for delivery, installation (except physical connection to facility utilities), startup, calibration, and training. Also included in this are four (4) bid options for annual on-site preventative maintenance, checks, services, and calibration. In their response, potential offerors should show how the proposed equipment meets or exceeds the salient characteristics listed below. Any exceptions or limitations to these requirements, if applicable, should also be identified.

SALIENT CHARACTERISTICS

1.) Physical Characteristics and Features

- a. The interior useable space of the chamber should have dimensions of at least 8 ft., 5 in. deep by 10 ft., 11 in. wide by 8 ft., 10 in. tall. Circulation systems and/or other components internal to the chamber should not obstruct or reduce this space.
- b. Offerors should indicate chamber floor height (from ground level) in their response.
- c. The chamber should be equipped with a double-door front entry.
 - i. The doors should have a handle or other device to allow easy opening/closure and be maneuverable by a single person.
 - ii. Each door should be equipped with a means for locking/clamping the door to engage door seals and maintain interior conditions.
 - iii. Each door should be equipped with a safety feature that prevents personnel from being locked inside of the chamber.
 - iv. The door and hinge system should be designed and constructed to allow the doors to be open for extended periods of time without damage.
 - v. Each door should open to a 180-degree angle, and there should be no obstruction to the entryway.
 - vi. Contractor should indicate height and width of doorway with both doors open in their response.

- vii. **Each door** should be equipped with a minimum of one (1) window for viewing the interior during testing. Offerors' response should include the specific location and dimensions of windows.
- d. The chamber exterior dimensions (as installed) should be provided in the response.
- e. The chamber should be equipped with lighting housed in durable, vapor-tight, corrosion resistant fixtures.
- f. The chamber floor should be heavy duty and capable of supporting heavy loads. Potential offerors should indicate floor load rating in pounds per square foot.
- g. All internal and external joints and seams should be securely seamed and vapor sealed to maintain test conditions.
- h. All materials used in the construction of the chamber should be selected to be compatible with the following test ladings: liquids (e.g., water, propylene glycol, etc.) and solids (e.g., steel, iron, aluminum, wood, plastic, etc.).
- i. The chamber should be equipped with a minimum of one (1) drain in the floor for handling condensate.
- j. The chamber should be equipped with a minimum of two (2), 4-inch diameter ports and plugs - one (1) on each side of the chamber.

2.) Performance Characteristics

- a. The chamber should provide a temperature range of at least -50 °F to +180 °F (-45 °C to +82 °C), with a control tolerance of +/- 2 °F (+/- 1.1 °C) or better, after stabilization.
- b. Offerors shall identify the cooling and heating rate in °F and/or °C per minute within the specified temperature range.
- c. The chamber should provide a humidity range of at least 20-95% Relative Humidity (RH) in dry bulb range of +68 °F to +180 °F (+20 °C to +82°C), with a control tolerance of +/- 5% RH.
- d. Deviations from the 20-95% RH range and/or limitations due to dry bulb temperatures or minimum dew points should be included in the response by the potential offeror.
- e. Potential offerors should identify optional dry air purge capability.

- f. The chamber should be capable of reaching and maintaining the rates, temperatures, and humidities specified above for up to six (6) months.
- g. The chamber should include circulation fans that provide sufficient volume and flow of air to obtain maximum uniformity of air temperature throughout the chamber.
- h. Potential offerors must include the chamber's performance characteristics in their response for the full temperature and humidity ranges specified above.

3.) Chamber Controls

- a. The chamber should be equipped with a digital, full color, touch screen controller.
- b. The controller should allow for separate test programs/profiles. Offerors should indicate the quantity of test profiles that may be saved.
- c. Offerors should specify the types of commands and test program types available within the controller.
- d. The controller should allow for the real-time monitoring of test parameters, test statuses, and graphical progress.
- e. The controller should provide data logging functionality with graphical charts and include the ability to archive temperature/humidity data and system logs.
- f. The controller should allow for the transfer of files and data onto and off of the controller utilizing USB 2.0 or higher. Data should be transferable in Microsoft Excel, comma separated values, or other common format.
- g. Offerors should indicate if the controller can be network-enabled for control and data acquisition.
- h. The controller should include on-screen diagnostics menus for preventative maintenance, calibration, and trouble shooting.
- i. The offeror should include an option for a self-printing circular chart recorder which tracks temperature and humidity, and provide both as printed chart data. It should also include digital indication of temperature and humidity values.

4.) Environmental Requirements

- a. The chamber should perform according to the performance specifications indicated in this document within a laboratory environment with a temperature range of 20 °C to 25.5 °C and a relative humidity of under 75%, non-

condensing.

- b. Chamber refrigerants should be CFC-free, environmentally safe, non-flammable, non-explosive, and have a Zero Ozone Depletion Potential (ODP).

5.) Utility Requirements

- a. In their response, potential offerors should specify all necessary facility specifications and auxiliary systems required to achieve the specified level of performance, including (but not limited to) power, water, and air.
- b. The chamber should use the existing power infrastructure (208 Volt, 3-phase, 200A, 60 Hertz supply).
- c. The chamber should include an electrical power disconnect to shut off power to the entire chamber.
- d. The chamber should include a feature which provides for recovery and/or restart after a power failure.
- e. The chamber should include a low water cut-off feature to automatically shut down the system in the event there is an interruption to the water supply to prevent damage.
- f. The chamber should include flowmeter(s) on the cooling system to monitor flow. Flowmeter(s) shall display measurements in gallons per minute (gpm).
- g. The chamber should be compatible with PSCC's closed-loop water recycling system, which may utilize water and/or a 50/50 propylene glycol and water mixture.
- h. Potential offerors should identify the the chambers maximum fluid consumption rate for the following scenarios:
 - i. Pure water that is 50-60 °F
 - ii. 50/50 propylene glycol/water mixture that is 75 °F
 - iii. 50/50 propylene glycol/water mixture that is 85 °F
- i. Potential offerors should identify input/return pipe diameters.
- j. Potential offerors should identify the location and required clearances for utility connections and chamber maintenance.
- k. Potential offerors should identify any filtration/treatment requirements.

DELIVERY

- 1.) If a contract is awarded as result of a future contract action, the Contractor will be responsible for delivery (including all costs) of the equipment to the PSCC testing facility located at Tobyhanna Army Depot, Building 2 - Bay 5, 11 Hap Arnold Boulevard, Tobyhanna, PA 18466.
- 2.) The shipment(s) should be capable of being delivered to PSCC's building and moved to the installation site within the main laboratory facility.
- 3.) Estimated building access clearance is 11' 11" wide and 12' tall. Final access clearances will be determined by the Contractor.

INSTALLATION, SETUP, AND CALIBRATION

- 1.) Potential offerors should provide in their response a listing of installation requirements to include utility, floor-space, foundation, and clearance.
- 2.) Potential offerors should be aware that any RFP that may follow will include a requirement for two (2) trips to PSCC to install, setup, calibrate, and test the system as well as provide hands-on training.
 - a. The Contractor will fully assembly the chamber, including integration of the controller, chart recorder, and flowmeter(s).
 - b. The Government will make all facility utility connections.
 - c. The Contractor will return to perform system startup, software/hardware setup, calibration, acceptance testing, and training. Calibration should be traceable to NIST standards.

ACCEPTANCE TESTING

Immediately following installation, setup, and calibration, the Contractor will demonstrate successful operation of the unit, including features/functionality listed within any subsequent contract

TRAINING

- 1.) Either following or concurrent with acceptance testing, the Contractor will provide basic operator training and equipment familiarization, to include:
 - a. Hands-on instruction and step-by-step instructions with references to applicable user manual information and test method standards listed herein.
 - b. Instruction regarding system maintenance and upkeep, acquainting system

users/maintainers with the preventative measures necessary to ensure optimum performance and maximum system life.

2.) Training will be provided for on-site personnel.

WARRANTY

Potential offerors should include their standard warranty terms & conditions in the response.

DOCUMENTATION/SOFTWARE

Potential offerors should explain any software requirements for using the device(s) or for transferring the data to a computer for analysis. Additionally, potential offerors are encouraged to elaborate on their device's adherence to any relevant industry or Government standard, as well as NIST traceability of measurement apparatuses.

OPTIONS

Potential offerors are encouraged to identify any maintenance and calibration recommendations for the device. The contemplated contract may include up to four (4) additional options for annual preventative maintenance, checks, services, and calibration. In their response, potential Contractors should include any pertinent information related to their ability and willingness to support this type of service arrangement – including whether that service would be performed at the Government's facility, the Contractor's facility, or through an authorized 3rd party.